CS251: Introduction to Language Processing

End Sem Exam Solutions

Question-1: 12 Marks

	Semantic Rules
S → E	print(E.code)
E → E1 + E2	E.code = E1.code || E2.code || '+'
E → E1 - E2	E.code = E1.code || E2.code || '-'
E → E1 * E2	E.code = E1.code || E2.code || '*'
E → (E1)	E.code = E1.code
E → Digit	E.code = Digit.lexval.

Question-2: 25 Marks

Part-(a): 15 Marks

S→ FOR (M1 E1; M2 E2; M3 E3) M4 S1
{
backpatch(E2.truelist, M4.quad);
backpatch(S1.nextlist, M3.quad);
S.nextlist = E2. falselist
E1.code || E2.code || E3. code ||
emit (goto M2.quad) || S1.code || emit (goto M3. quad)
} //14 Marks.

M → ∈ {M.quad = nextquad } //1 Mark

Part-(b): 10 Marks

100: i=0
101: t1=x+y
102: if i<t1 goto 107
103: goto 111
104: t2=i+1
105: i=t2
106: goto 101
107: t3=i*x
108: t4=t3+y
109: i=t4
110: goto 104
111:

Question-3:

Part-(a): 3 Marks for correct algorithm

Labelling algorithm:

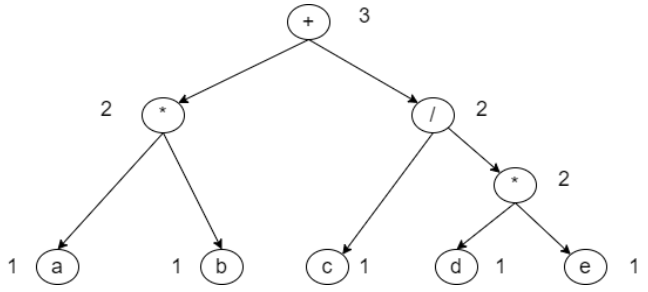
Since both the operands of the binary operation must be registers, we require 1 register for each leaf node to load the contents of the memory location to the register. Hence the modified algorithm is as follows.

1. **Label each leaf node by 1**
2. If the labels of the children of a node n are l_1 and l_2 then

$$\begin{aligned} \text{label}(l) &= \max(l_1, l_2), \quad \text{if } l_1 \neq l_2 \\ &= l_1+1 \qquad\quad \text{Otherwise} \end{aligned}$$

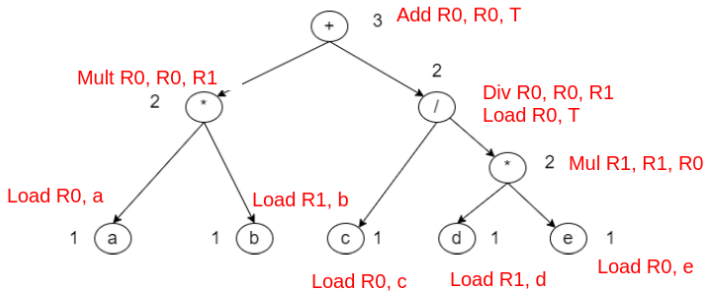
Part-(b): 5 Marks for the correct labelled tree.

The labelled expression tree is shown below.



The minimum number of registers required to compute the expression a*b+c/(d*e) is **3**.

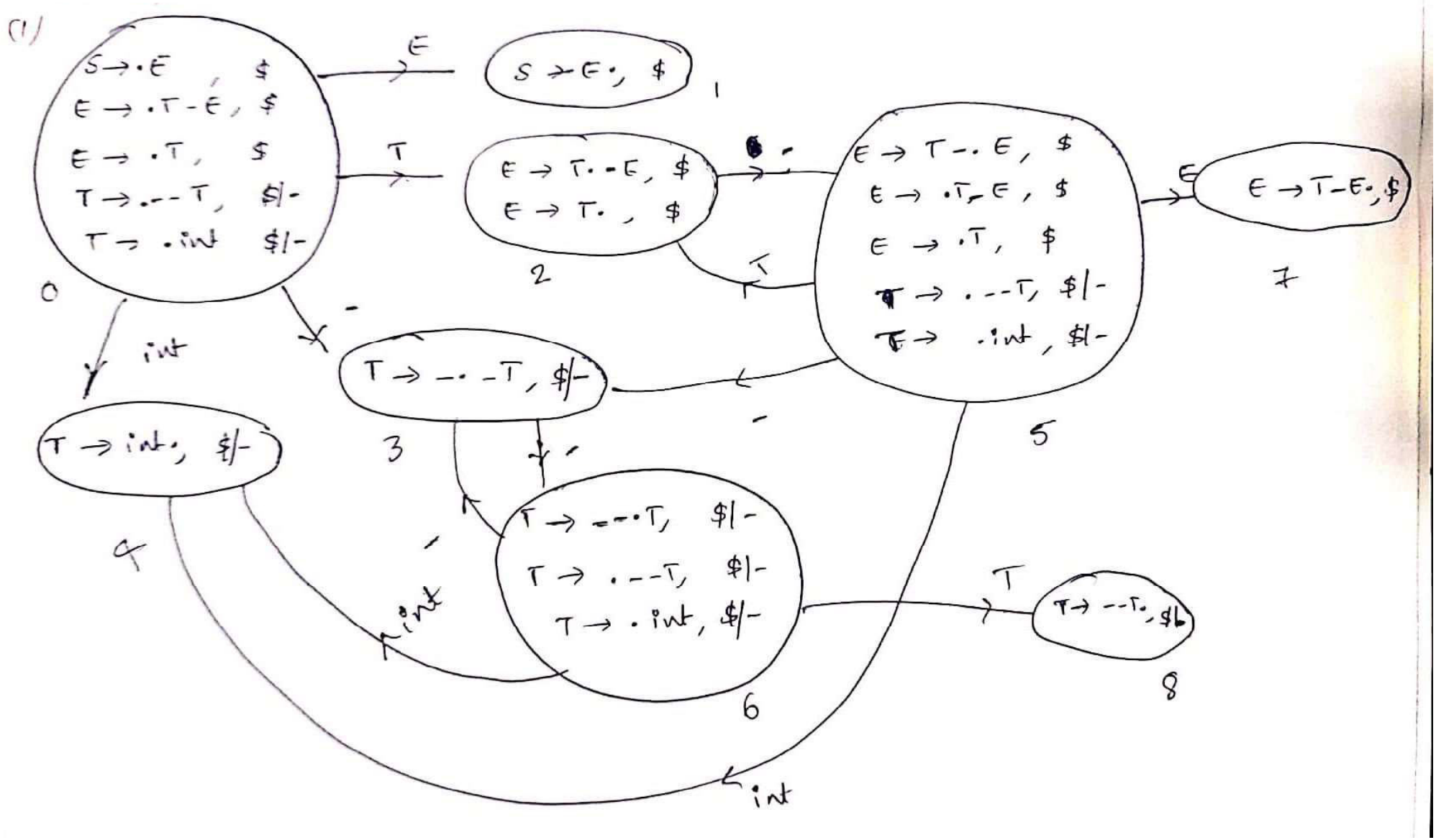
Part-(c): 15 Marks.



Sequence of assembly instructions. Each correct instruction carries 1.5 Marks (except the last two which carry 0.75 marks each).The sequence should be correct in the order specified to get the required marks.

```
Load R1, d
Load R0, e
Mul R1, R1, R0
Load R0, c
Div R0, R0, R1
Load R0, T
Load R0, a
Load R1, b
Mul R0, R0, R1
Load R1, T
Add R0, R0, R1
```

Question-4 (1) (Carries 13.5 Marks. Each correct state fetches 1.5 Marks)



Scanned with CamScanner

Question-4 (2): Carries 10 Marks. Each correct entry in the table fetches 0.5 Marks

	-	int	$	E	T
0	S3	S4		1	2
1			acc		
2	S5		r3		
3	S6				
4		r5	r5		
5	S3	S4		7	2
6	S3	S4			8
7			r2		
8	r4		r4		

S → E (1)
E → T - E (2)
E → T (3)
T → - - T (4)
T → int (5)

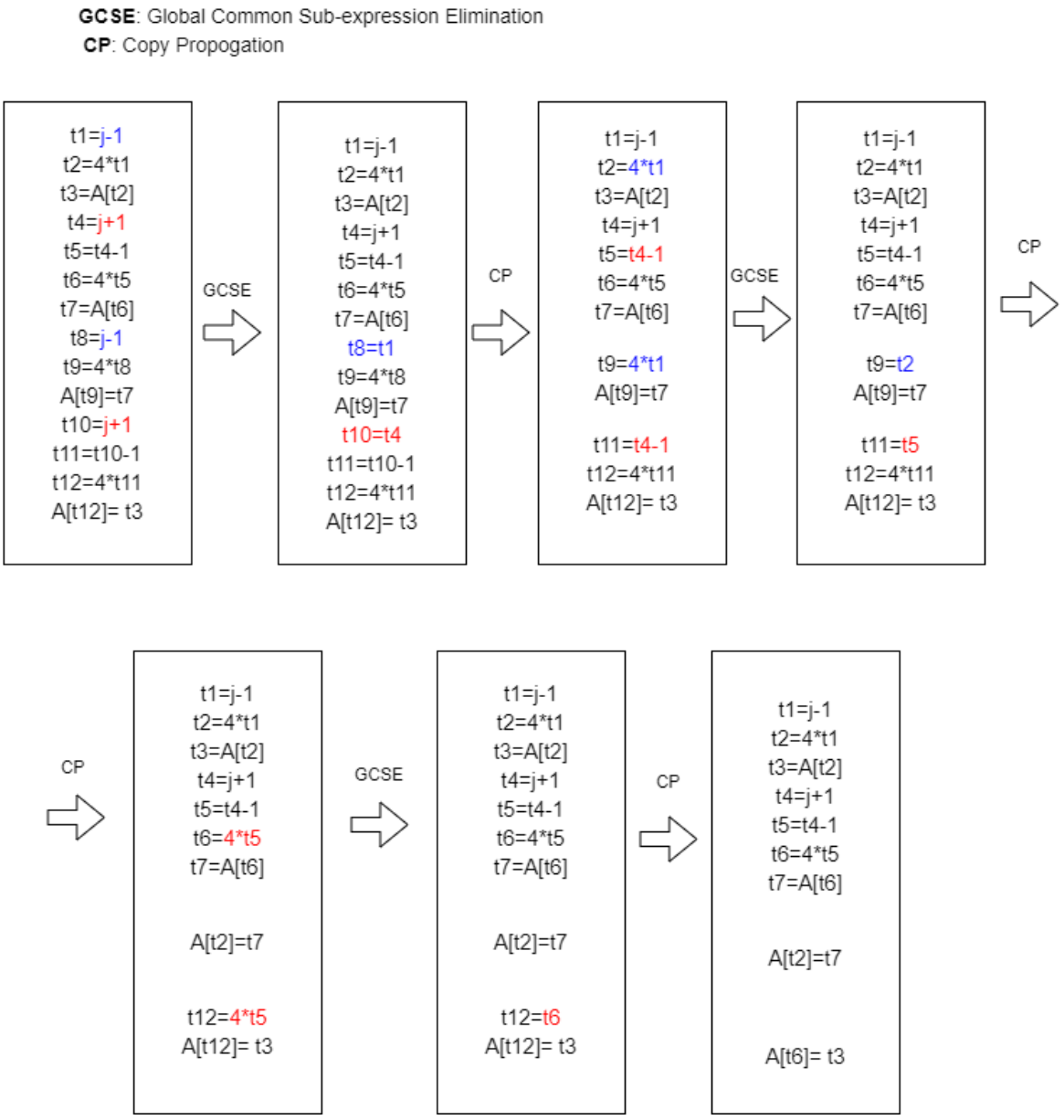
Question-4 (3): The correct justification will fetch you 1.5 Marks

The grammar is LR(1) because each cell in the LR(1) parser table contains atmost one entry.

Scanned with CamScanner

Question-5: 15 Marks.

There are 10 optimizations (2 GCSE+2 CP+2 GCSE+ 2 CP+1 GCSE+1CP) are applied (highlighted in color). Each one carries 1.5 marks.



There is no dead code in the final three-address code generated after applying GCSE and CP; hence dead-code elimination is not applicable.